FOR A SURJECTIVE CLOSED CONTINUOUS MAP THE CLOSED SETS, OPENS AND LOCALLY CLOSED SETS ARE EXACTLY THE PREIMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter, staged here as sources/stacks-topology/preamble.tex; this card uses retained-preamble.tex, which copies that shared preamble mechanically, including its reference/slogan/history display environments. Nothing was added, removed or redefined; the wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: the chapter carries no chapter-local bibliography and neither the counted statement nor its proof contains a citation, so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: the card was taken from the staged candidate list for this chapter, candidate lemma-closed-morphism-quotient-topology; the statement and proof spans are that candidate's own statement_lines and proof_lines, checked against the pinned chapter so that each span opens and closes a real, uncommented environment.

Allowed standard prerequisites (editorial).

- The behaviour of closures under a continuous map and closedness of the map, together with the notion of submersiveness. Set up in the author's section on submersive maps, immediately before the counted statement. Fixes the four assertions and the closing claim that the map is submersive.

Lemma 0.1. *Let $f : X \to Y$ be surjective, closed, continuous map of topological spaces. Let $T \subset Y$ be a subset. Then*

(1) $\overline{T} = f(\overline{f^{-1}(T)})$,
(2) *$T \subset Y$ is closed if and only if $f^{-1}(T)$ is closed,*
(3) *$T \subset Y$ is open if and only if $f^{-1}(T)$ is open, and*
(4) *$T \subset Y$ is locally closed if and only if $f^{-1}(T)$ is locally closed.*

In particular we see that f is submersive.

Proof. It is clear that $\overline{f^{-1}(T)} \subset f^{-1}(\overline{T})$. Then $T \subset f(\overline{f^{-1}(T)}) \subset \overline{T}$ is a closed subset, hence we get (1). Part (2) is obvious from the fact that f is closed and surjective. Part (3) follows from (2) applied to the complement of T. For (4), if $f^{-1}(T)$ is locally closed, then $f^{-1}(T) \subset \overline{f^{-1}(T)}$ is open. Since the map $\overline{f^{-1}(T)} \to \overline{T}$ is surjective by (1) we can apply part (3) to the map $\overline{f^{-1}(T)} \to \overline{T}$ induced by f to conclude that T is open in $\overline{T}$, i.e., T is locally closed. □

Date: stacks-project:topology.tex:local-snapshot-20261012; CC BY 4.0.